Remote sensing-based detection and tracking of marine litter (special emphasis to macro-plastic) along the Indian coast

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Outline

Experiment 1 - Hyperspectral spectrometer study

- Experiment design and data collection
- Models and results

Experiment 2 - Multispectral (410-940 nm) study

- Implemented plan (SMOTE, NDVI, FDI, LDA)
- Models and results

Experiment 3 - Sentinel-2 satellite imagery (latest work)

- MADOS dataset, indices, Sea Snot Index
- DBSCAN, feature selection
- Image-level accuracy: Closing / Modulo / SAM
- Site testing: Limassol Bay, Cooum River, Interceptors
- Future work

Experiment 1

Hyperspectral spectrometer study

Experiment Design

- Description of collected data:

Object Class	Venue	Count	Label
Water	Pool	30	0
Floating Plastic	Pool	200	1
Floating Plastic	Bopal Canel	62	1
Vegetation	Outdoors (Nirma Campus)	31	0
Floating Vegetation	Outdoors (Nirma Campus)	30	0
Soil	Outdoors (Nirma Campus)	30	0
Iron	Outdoors (Nirma Campus)	30	0
Concrete	Outdoors (Nirma Campus)	25	0
Sediment Water	Pool	31	0

Table: Data Description

- Total Data Points: 469

Results

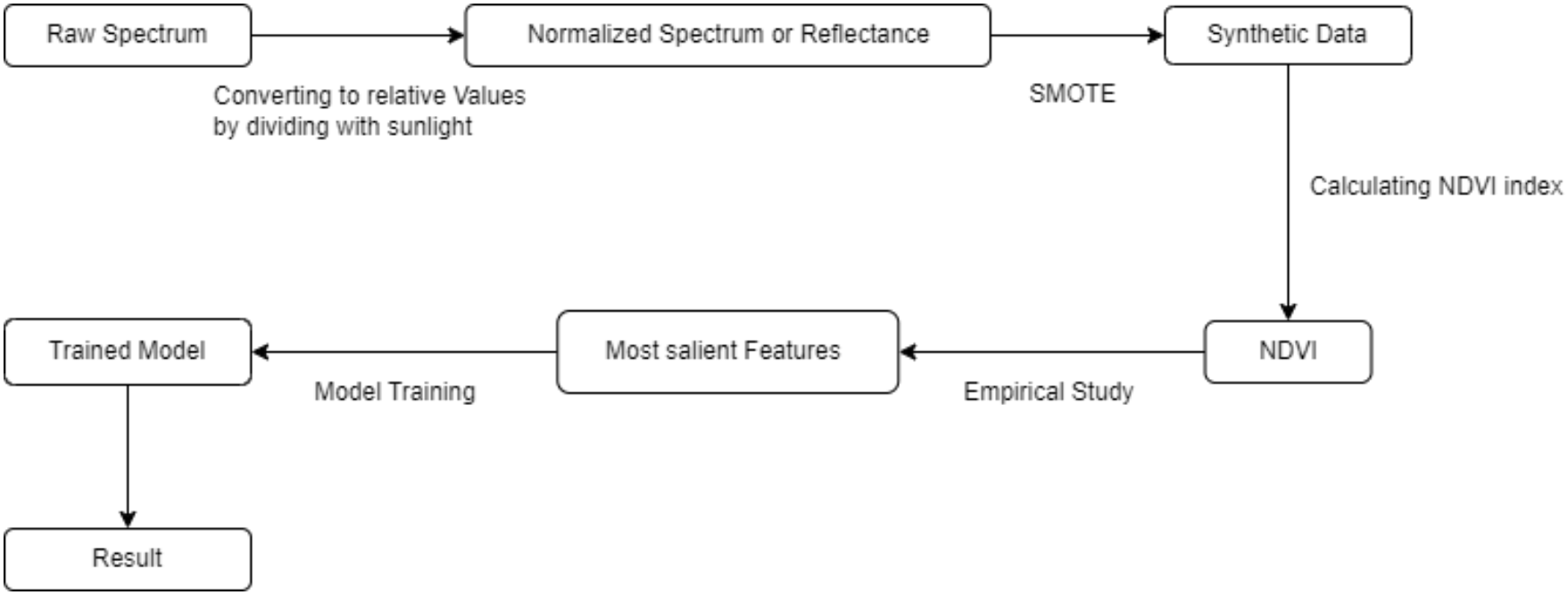
Model	Over All Accuracy	Accuracy on 1% Pixel Coverage	Accuracy on SAC Data
Random Forest	0.9	1	0.7
1D CNN + Transformer	0.97	1	1
1D CNN	0.84	0.98	0.58
SVM	0.96	0.98	0.68
Trained SVM	0.5	0	0.06

Experiment 2

Multispectral (410-940 nm) study

Implemented Plan

Below Figure shows implemented plan:



Models applied on data

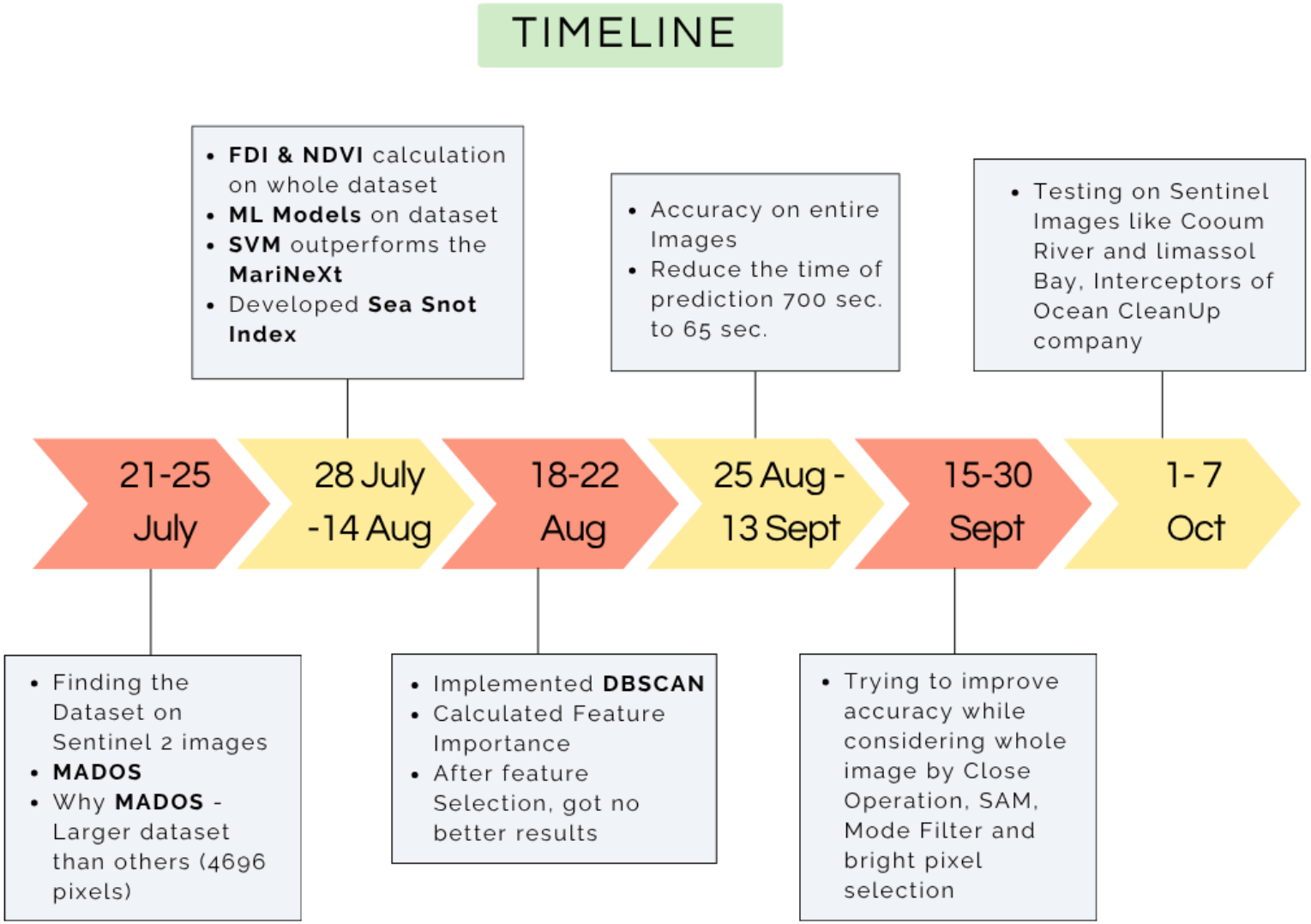
Models	**Input**	**epochs**	**Accuracy**
Random Forest	All Bands, NDVI	-	0.98
Multilayer Perceptron		30	1
SVM	All Bands, NDVI	-	1
kNN (IBk) (5)	All Bands, NDVI	-	0.97
Decision Tree	All Bands, NDVI	-	0.97
Gaussian NB	All Bands, NDVI	-	0.95
AdaBoost	All Bands, NDVI	-	0.99
Self supervised Learning	All Bands, NDVI	50 - 100	0.87 - 0.96

Table: Models & Accuracy

Experiment 3

Sentinel-2 satellite imagery (latest)

Timeline


TIMELINE
21-25 July
28 July -14 Aug
18-22 Aug
25 Aug - 13 Sept
15-30 Sept
1- 7 Oct
• Finding the Dataset on Sentinel 2 images
• MADOS
• Why MADOS - Larger dataset than others (4696 pixels)
• FDI & NDVI calculation on whole dataset
• ML Models on dataset
• SVM outperforms the MariNeXt
• Developed Sea Snot Index
• Implemented DBSCAN
• Calculated Feature Importance
• After feature Selection, got no better results
• Accuracy on entire Images
• Reduce the time of prediction 700 sec. to 65 sec.
• Trying to improve accuracy while considering whole image by Close Operation, SAM, Mode Filter and bright pixel selection
• Testing on Sentinel Images like Cooum River and limassol Bay, Interceptors of Ocean CleanUp company

- Datasets Available On Sentinel-2

Sr. No.	Dataset	Preprocessing	No. of bands
1	MARIDA [1]	ACOLITE	11
2	MADOS [2]	ACOLITE	11
3	MAIA-S2 [3]	Not Required (Less Height)	9
4	FDI [4]	ACOLITE	13
5	Plastic Object [5]	ACOLITE	13
6	Bay Islands of Honduras [6]	ACOLITE	13
7	ISPRS Dataset [7]		13
8	Manila Bay [8]	ACOLITE	13

- Description of MADOS Dataset
  - 2803 images (240 X 240)
  - 15 lacs annotated pixels
  - 15 Classes
  - 4696 annotated as Marine Debris
  - 3.6 lacs test pixels
  - 7.6 lacs train pixels
  - 3.5 lacs validation pixels
  - 722 test images

- Models trained on MADOS
- The best results were achieved using only 11 spectral bands

Model	MD			Classes
	Recall	Precision	F1-Score	
SVM	0.83	0.6	**0.7**	15
	No better Results			7
Random Forest	0.68	0.43	0.53	15
KNN	0.77	0.5	0.61	15
Logistic Regression	0.16	0.52	0.24	15
Naive Bayes	Poor Performance			15
LDA	0.31	0.23	0.26	15

Table: Performance of Different Models

28 July - 14 August

- FDI and NDVI on Dataset

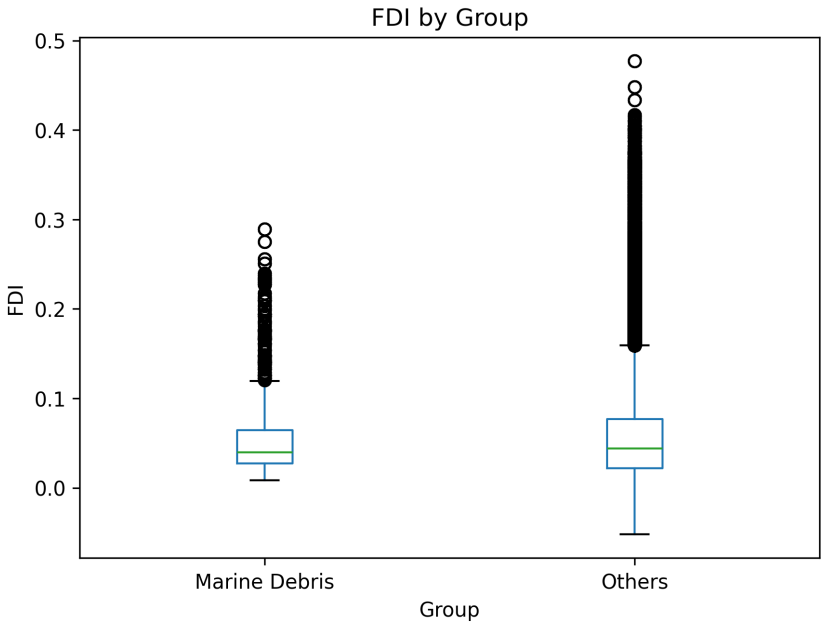


Figure: Floating Debris Index

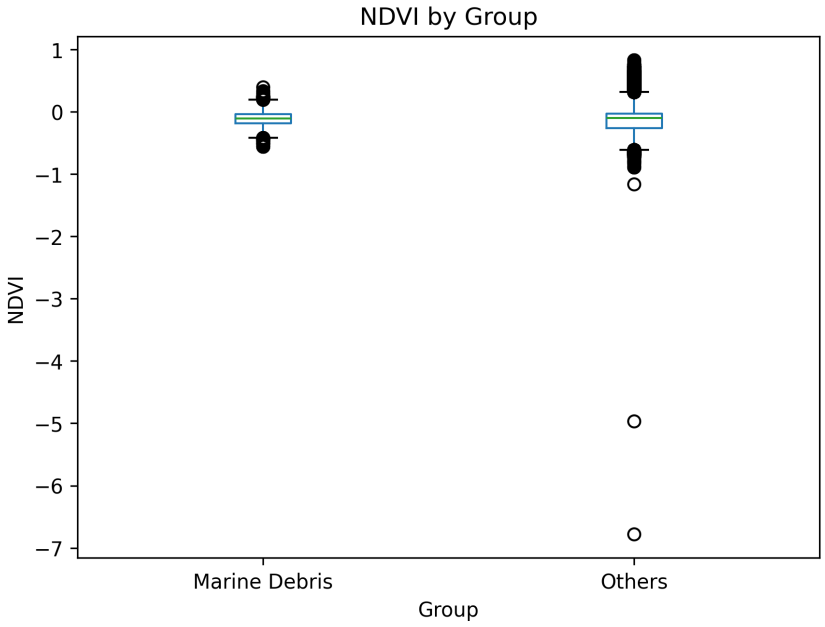


Figure: NDVI

- Conclusion from Index calculation
  - Only FDI & NDVI cannot determine the pixel class
  - Their inclusion during training did not improve model performance.

28 July - 14 August

- Similar to the NDVI index concept, the significant spectral gap observed in sea snot, can be effectively utilized for its detection and analysis.

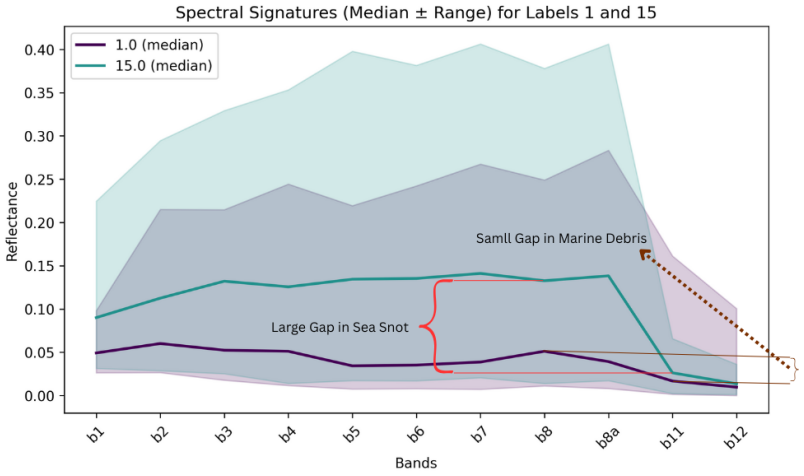


Figure: Spectrum of Marine Debris and Sea Snot

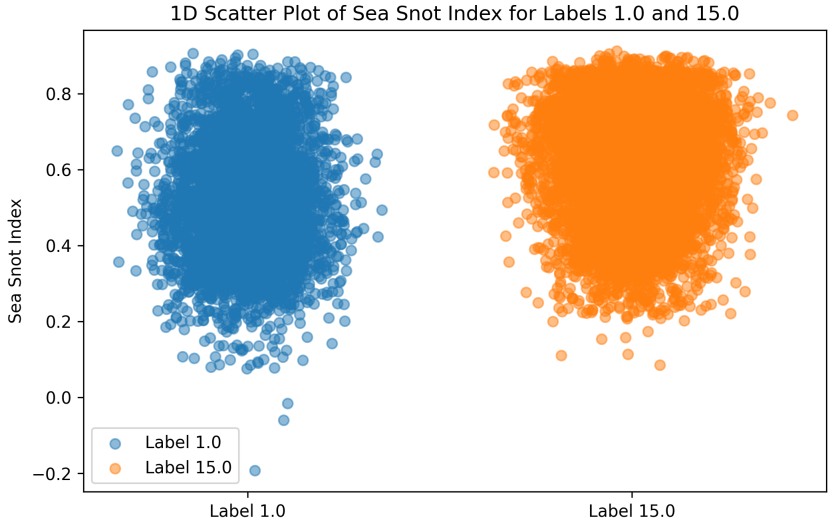
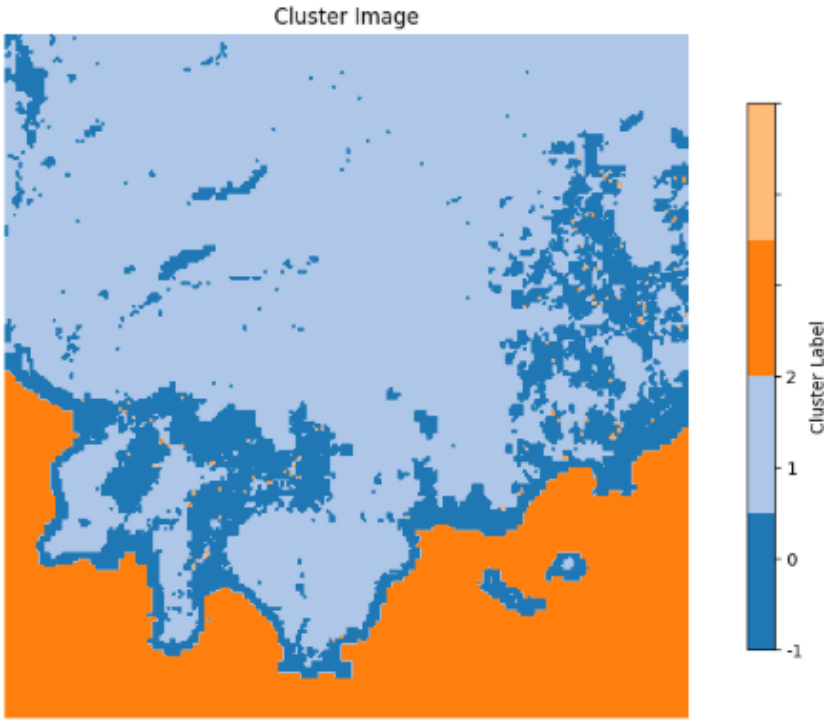


Figure: Sea Snot Index

- Formula of Sea Snot Index
  - Sea Snot Index = $\frac{R_{\text{Band 8}} - R_{\text{Band 11}}}{R_{\text{Band 8}} + R_{\text{Band 11}}}$
- Training with that index did not yield better results.

- Results of DBSCAN



- Drawbacks with DBSCAN
  - The algorithm produces a varying number of clusters across different images, even with the same parameter settings. (eps, min_samples)
  - The resulting cluster labels are often difficult to interpret meaningfully.

- Features Selection & Traning
  - Feature selection was done using the Mutual Information
  - The threshold is set to 1.5 to select the most important features, and the remaining ones are dropped
  - The selected features did not lead to good results

```
b1: 1.6837051945285395
b2: 0.621969401519844
b3: 0.8572582157465785
b4: 1.0273996294828895
b5: 1.5722292853810775
b6: 1.5675738505881767
b7: 1.572052497710367
b8: 1.1966109323660383
b8a: 1.5735236640857702
b11: 1.6212884077200658
b12: 1.6223004288068987
fdi: 1.5393347819029872
ndvi: 0.4215744448382086
```

Figure: Mutual Information of Each Features

25 August - 13 September

- Labels
  0. Marine Debris
  1. Dense Sargassum
  2. Sparse Floating Algae
  3. Natural Organic Material
  4. Ship
  5. Oil Spill
  6. Marine Water
  7. Sediment-Laden Water
  8. Foam
  9. Turbid Water
  10. Shallow Water
  11. Waves Wakes
  12. Oil Platform
  13. Jellyfish
  14. Sea snot

- The output of the SVM clearly indicates that the location of the garbage is accurately identified, with no significant pixel-level discrepancies.
- Now, let's consider this as an image classification problem, which provides a more realistic scenario. It is similar to a deployed model where an image is provided for plastic detection.
- Here, we are only considering MD class.

TP	FP	TN	FN	Recall	Precision	F1-Score
121	194	405	2	0.98	0.38	0.55

Table: SVM Performance on Entire Images Across the Whole Dataset

- Reason behind False Positive images
  - Model is predicting MD pixel in object like ship, island

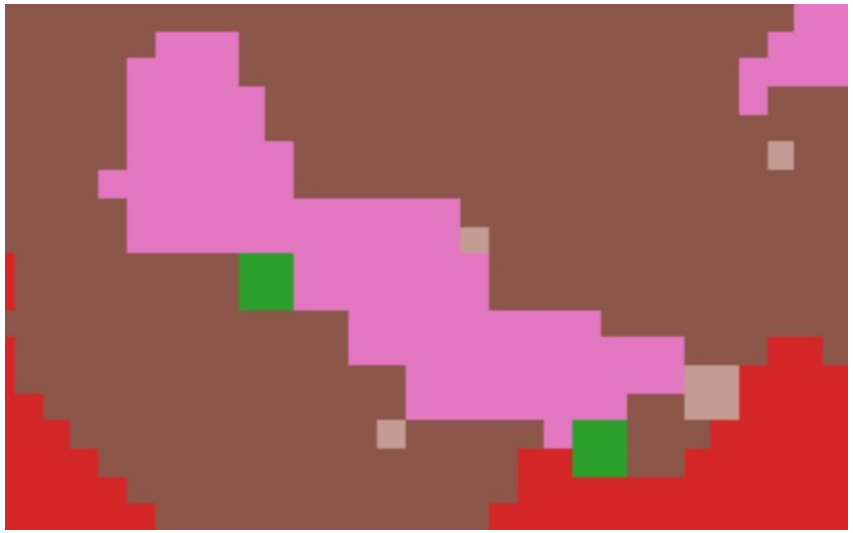
Figure: MD pixel in Island

15-30 September

- Enhancing image-level accuracy
  1. Closing Operation
  2. Modulo Filter
  3. Segment Anything Model (SAM)
  4. Bright Pixel Selection (Yet to implement)

- Result of Closing Operation

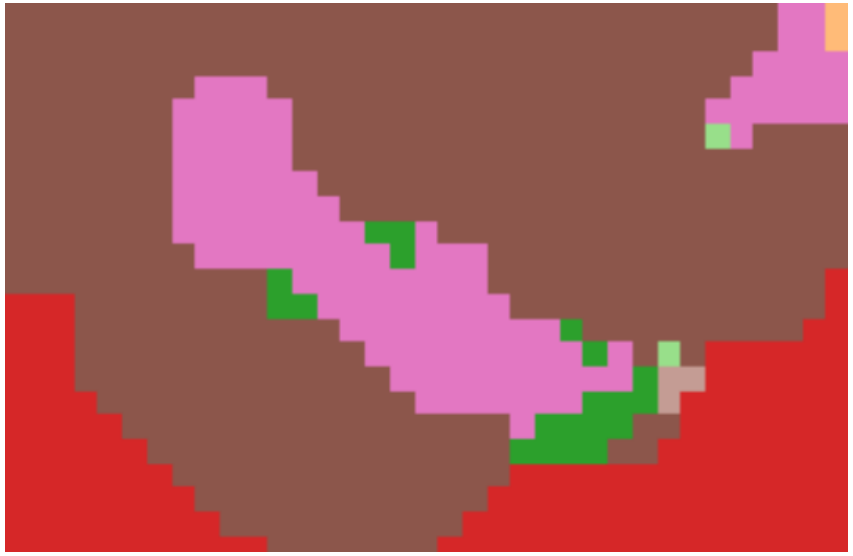
(a) Before applying Closing Operation



(b) After applying Closing Operation

- Result of Modulo Filter

(a) Before applying Modulo Filter



(b) After applying Modulo Filter

3. Segment Anything Model (SAM)

- SAM model is developed by Meta
- This model can segment any object based on the parameters set, such as the minimum number of pixels
- It separates each object or floating object and assigns a distinct label to each

Figure: SAM Model Output

- Results of all the techniques

Technique	Recall	Precision	F1-Score	Filter Size
SVM	0.98	0.38	0.55	-
Closing Operation	0.05	0.31	0.09	3x3
	0.48	0.49	0.48	2x2
Modulo Filter	0.69	0.42	0.52	3x3
SAM	0.45	0.48	0.47	-

- As we aimed to improve the F1 score, we observed a compromise in the recall that had been previously achieved. Therefore, we decided to test the model on Sentinel-2 images

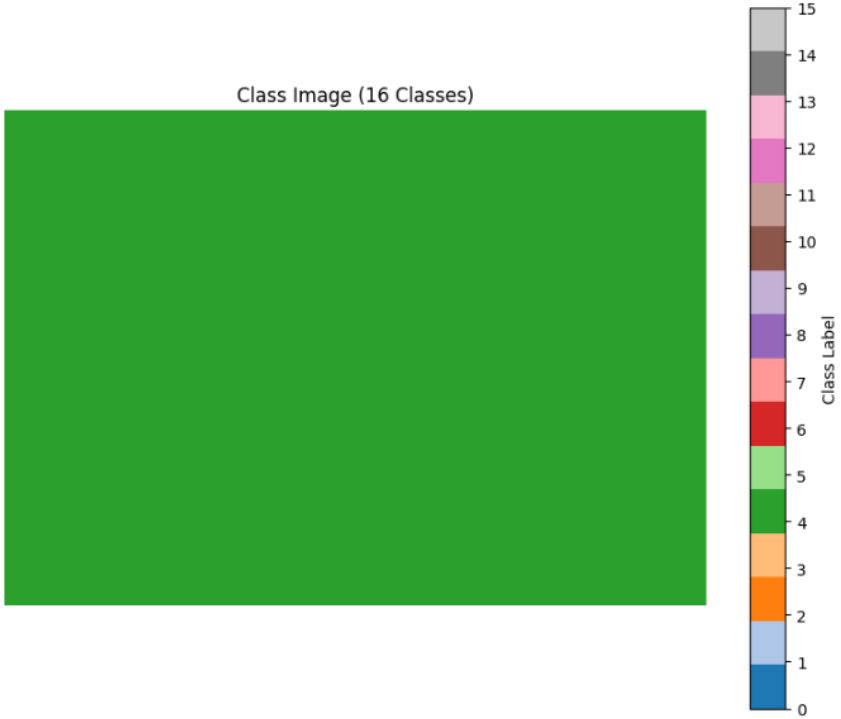
1-7 October

- Testing on sites
  1. Limssol Bay
  2. Cooum River
  3. Interceptors of Ocean CleanUp company

1-7 October

1. Limssol Bay

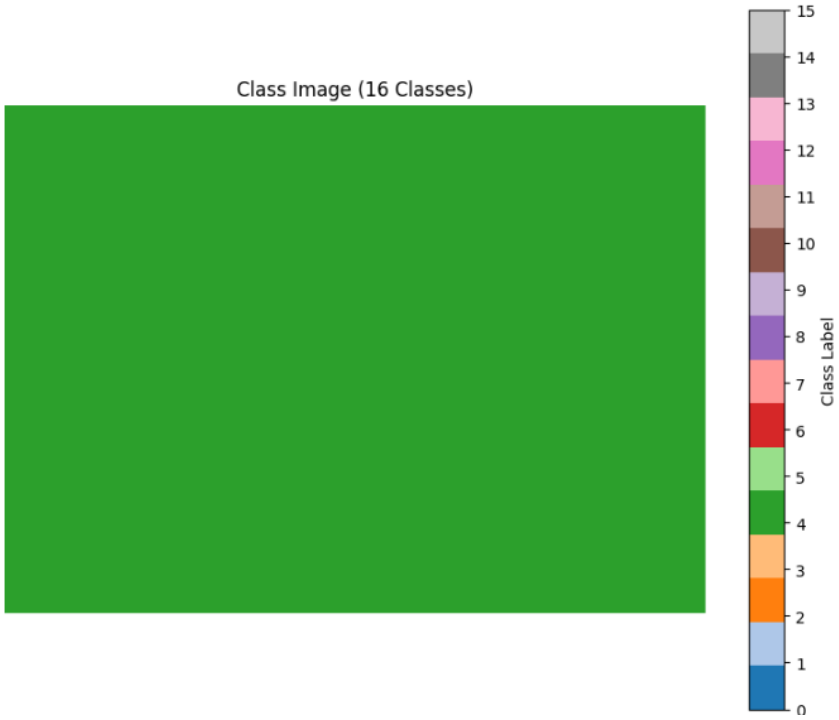
(a) RGB Image



(b) Segmented Image

2. Cooum River

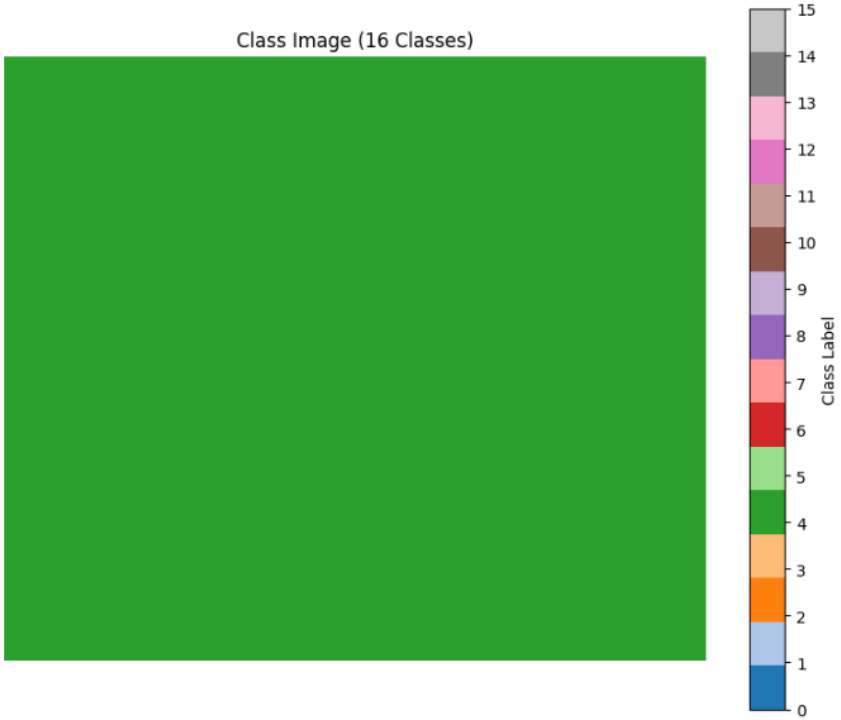
(a) RGB Image



(b) Segmented Image

3. Interceptors of Ocean CleanUp companyy

(a) RGB Image



(b) Segmented Image

1-7 October

- Reason for the suboptimal results on downloaded Sentinel-2 images:
  - Atmospheric correction using ACOLITE was not applied prior to testing.
  - I tried running the ACOLITE processor but encountered a 'NoneType Error'

Future Work

- Apply ACOLITE preprocessing to the images.
- If satisfactory results are not achieved after applying ACOLITE, enhance the model accordingly.

Thank You